Indice

Dataset

The dataset contains 506 observations from the United States of America Bureau of census, regarding real estate housing in Boston, Massachussets. Data originally was used by Harrison, D and Rubinfeld, D.L. in ‘Hedonic prices and the demand for clean air’, J. Environ, Economics & Management, vol. 5, 81-102, 1978.

The dataset source is the following:
https://www.kaggle.com/datasets/avish5787/boston-data-set/data

Variables

CRIM - per capita crime rate by town

ZN - proportion of residential land zoned for lots over 25,000 sq.ft.

INDUS - proportion of non-retail business acres per town.

CHAS - Charles River dummy variable (1 if tract bounds river; 0 otherwise)

NOX - nitric oxides concentration (parts per 10 million)

RM - average number of rooms per dwelling

AGE - proportion of owner-occupied units built prior to 1940

DIS - weighted distances to five Boston employment centres

RAD - index of accessibility to radial highways

TAX - full-value property-tax rate per $10,000

PTRATIO - pupil-teacher ratio by town

B - 1000(Bk - 0.63)^2 where Bk is the proportion of blacks by town

LSTAT - % lower status of the population

MEDV - Median value of owner-occupied homes in $1000's

Data cleaning

The dataset has no missing value, as shown from the following table

Statistiche univariata

	N	Media	Deviazione std.	Mancante		N. di estremi[a]	
				Conteggio	Percentuale	Basso	Alto
Price	506	22,533	9,1971	0	,0	2	38
CRIM	506	3,6135236	8,60154511	0	,0	0	66
ZN	506	11,364	23,3225	0	,0	0	68
INDUS	506	11,1368	6,86035	0	,0	0	0
NOX	506	,554695	,1158777	0	,0	0	0
RM	506	6,28463	,702617	0	,0	8	22
AGE	506	68,575	28,1489	0	,0	0	0
DIS	506	3,795043	2,1057101	0	,0	0	5
RAD	506	9,549	8,7073	0	,0	0	0
TAX	506	408,237	168,5371	0	,0	0	0
PTRATIO	506	18,456	2,1649	0	,0	15	0
B	506	356,6740	91,29486	0	,0	76	0
LSTAT	506	12,6531	7,14106	0	,0	0	6
CHAS	506			0	,0		

a. Numero di casi fuori dall'intervallo (Q1 - 1.5*IQR, Q3 + 1.5*IQR).

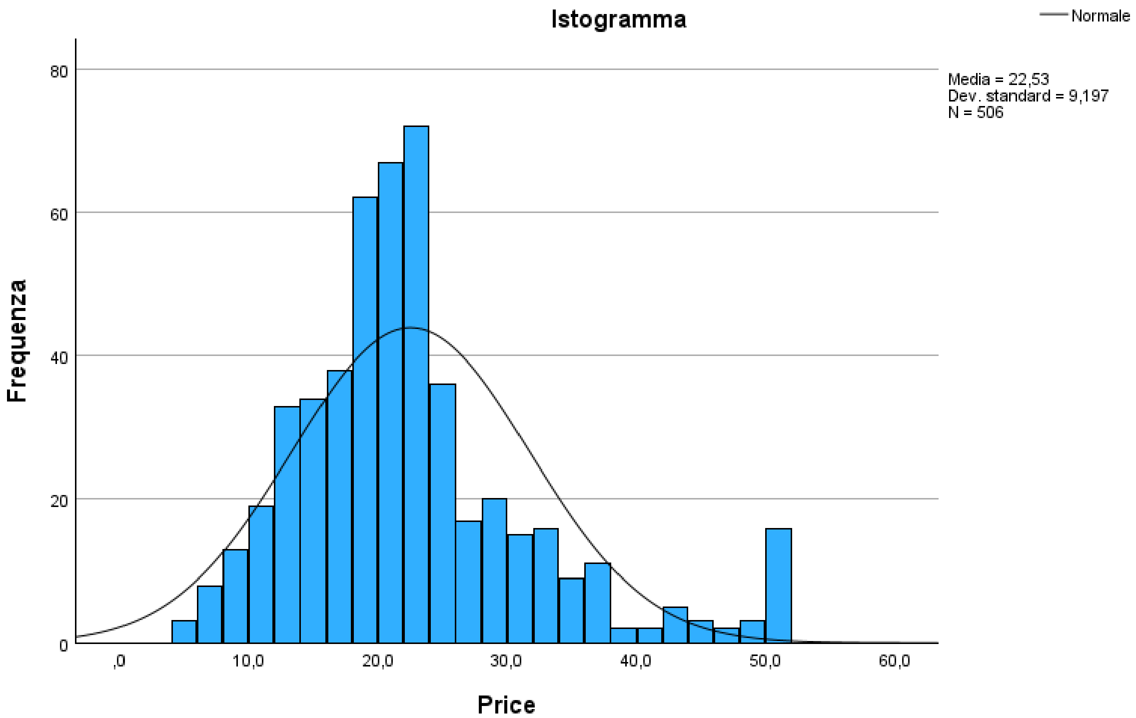
Descriptives on price

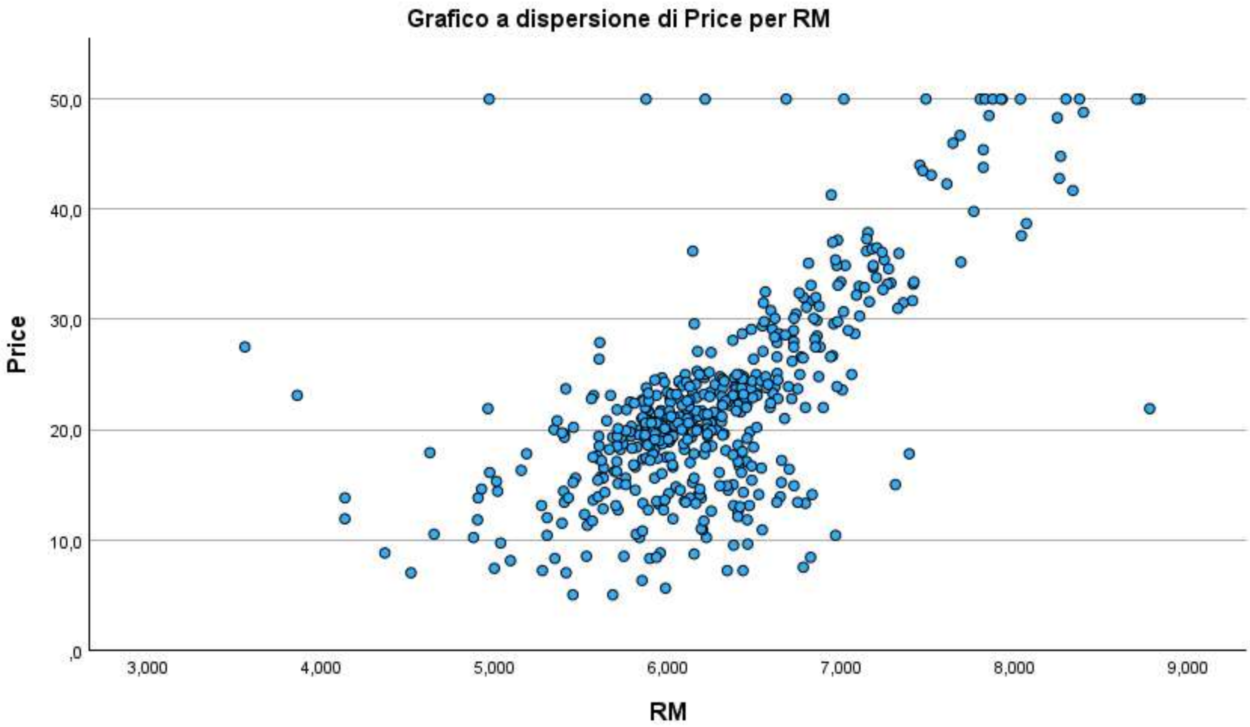
Statistiche descrittive

	N	Minimo	Massimo	Media	Deviazione std.	Varianza	Asimmetria		Curtosi	
	Statistica	Statistica	Statistica	Statistica	Statistica	Statistica	Statistica	Errore standard	Statistica	Errore standard
Price	506	5,0	50,0	22,533	9,1971	84,587	1,108	,109	1,495	,217
Numero di casi validi (listwise)	506									

House prices have a mean of 22,5 thousand of dollars, and a standard deviation of 9,1971.

Mover, there's positive asymmetry (1,108) and the distribution presents kurtosis (leptokurtosis: 1,495)





House prices over 50 000$ are censored at 50 000$, as shown in the graph.

From here the anomaly of the normal distribution in the preceding page.

Collinearity diagnostics

Correlazioni

		CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT
CRIM	Correlazione di Pearson	1	-,200**	,407**	-,056	,421**	-,219**	,353**	-,380**	,626**	,583**	,290**	-,385**	,456**
	Sign. (a due code)		<,001	<,001	,209	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
ZN	Correlazione di Pearson	-,200**	1	-,534**	-,043	-,517**	,312**	-,570**	,664**	-,312**	-,315**	-,392**	,176**	-,413**
	Sign. (a due code)	<,001		<,001	,338	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
INDUS	Correlazione di Pearson	,407**	-,534**	1	,063	,764**	-,392**	,645**	-,708**	,595**	,721**	,383**	-,357**	,604**
	Sign. (a due code)	<,001	<,001		,157	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
CHAS	Correlazione di Pearson	-,056	-,043	,063	1	,091*	,091*	,087	-,099*	-,007	-,036	-,122**	,049	-,054
	Sign. (a due code)	,209	,338	,157		,040	,040	,052	,026	,869	,424	,006	,273	,226
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
NOX	Correlazione di Pearson	,421**	-,517**	,764**	,091*	1	-,302**	,731**	-,769**	,611**	,668**	,189**	-,380**	,591**
	Sign. (a due code)	<,001	<,001	<,001	,040		<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
RM	Correlazione di Pearson	-,219**	,312**	-,392**	,091*	-,302**	1	-,240**	,205**	-,210**	-,292**	-,356**	,128**	-,614**
	Sign. (a due code)	<,001	<,001	<,001	,040	<,001		<,001	<,001	<,001	<,001	<,001	,004	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
AGE	Correlazione di Pearson	,353**	-,570**	,645**	,087	,731**	-,240**	1	-,748**	,456**	,506**	,262**	-,274**	,602**
	Sign. (a due code)	<,001	<,001	<,001	,052	<,001	<,001		<,001	<,001	<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
DIS	Correlazione di Pearson	-,380**	,664**	-,708**	-,099*	-,769**	,205**	-,748**	1	-,495**	-,534**	-,232**	,292**	-,497**
	Sign. (a due code)	<,001	<,001	<,001	,026	<,001	<,001	<,001		<,001	<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
RAD	Correlazione di Pearson	,626**	-,312**	,595**	-,007	,611**	-,210**	,456**	-,495**	1	,910**	,465**	-,444**	,489**
	Sign. (a due code)	<,001	<,001	<,001	,869	<,001	<,001	<,001	<,001		<,001	<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
TAX	Correlazione di Pearson	,583**	-,315**	,721**	-,036	,668**	-,292**	,506**	-,534**	,910**	1	,461**	-,442**	,544**
	Sign. (a due code)	<,001	<,001	<,001	,424	<,001	<,001	<,001	<,001	<,001		<,001	<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
PTRATIO	Correlazione di Pearson	,290**	-,392**	,383**	-,122**	,189**	-,356**	,262**	-,232**	,465**	,461**	1	-,177**	,374**
	Sign. (a due code)	<,001	<,001	<,001	,006	<,001	<,001	<,001	<,001	<,001	<,001		<,001	<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
B	Correlazione di Pearson	-,385**	,176**	-,357**	,049	-,380**	,128**	-,274**	,292**	-,444**	-,442**	-,177**	1	-,366**
	Sign. (a due code)	<,001	<,001	<,001	,273	<,001	,004	<,001	<,001	<,001	<,001	<,001		<,001
	N	506	506	506	506	506	506	506	506	506	506	506	506	506
LSTAT	Correlazione di Pearson	,456**	-,413**	,604**	-,054	,591**	-,614**	,602**	-,497**	,489**	,544**	,374**	-,366**	1
	Sign. (a due code)	<,001	<,001	<,001	,226	<,001	<,001	<,001	<,001	<,001	<,001	<,001	<,001	
	N	506	506	506	506	506	506	506	506	506	506	506	506	506

**. La correlazione è significativa a livello 0,01 (a due code).

*. La correlazione è significativa a livello 0,05 (a due code).

As the Pearson correlation table demonstrates, the linear positive relationship between the explicative variables TAX and RAD is almost perfect, specifically, is 0,91.

In the following table, it's noted that variables with the highest correlation have also highest VIF, near 10. Particularly, TAX has a 9 VIF, whereas RAD is 7,5.

Coefficienti[a]

Modello		Coefficienti non standardizzati		Coefficienti standardizzati			Statistiche di collinearità	
		B	Errore standard	Beta	t	Sign.	Tolleranza	VIF
1	(Costante)	36,459	5,103		7,144	<,001		
	CRIM	-,108	,033	-,101	-3,287	,001	,558	1,792
	ZN	,046	,014	,118	3,382	<,001	,435	2,299
	INDUS	,021	,061	,015	,334	,738	,251	3,992
	CHAS	2,687	,862	,074	3,118	,002	,931	1,074
	NOX	-17,767	3,820	-,224	-4,651	<,001	,228	4,394
	RM	3,810	,418	,291	9,116	<,001	,517	1,934
	AGE	,001	,013	,002	,052	,958	,322	3,101
	DIS	-1,476	,199	-,338	-7,398	<,001	,253	3,956
	RAD	,306	,066	,290	4,613	<,001	,134	7,484
	TAX	-,012	,004	-,226	-3,280	,001	,111	9,009
	PTRATIO	-,953	,131	-,224	-7,283	<,001	,556	1,799
	B	,009	,003	,092	3,467	<,001	,742	1,349
	LSTAT	-,525	,051	-,407	-10,347	<,001	,340	2,941

a. Variabile dipendente: Price

To exclude almost collinear regressors, I decided to eliminate the explicative variable with the highest VIS, near 10, in the general model.

Models

The objective of our analysis is to build a linear multiple regression model capable of explaining the relationship between the house prices variable and other descriptive variables.

I used the IBM SPSS software.

The first task at hand is to present the complete regressin model, including every selected variable, except for TAX.

Results of the first model (complete)

As we can see from the executed testing, every independent variable was put into the model, except for TAX.

Variabili immesse/rimosse[a]

Modello	Variabili immesse	Variabili rimosse	Metodo
1	LSTAT, CHAS, B, PTRATIO, ZN, CRIM, RM, INDUS, AGE, RAD, DIS, NOX[b]	.	Inserisci

a. Variabile dipendente: Price

b. Sono state immesse tutte le variabili richieste.

Riepilogo del modello[b]

Modello	R	R-quadrato	R-quadrato adattato	Errore std. della stima
1	,857[a]	,735	,729	4,7920

a. Predittori: (costante), LSTAT, CHAS, B, PTRATIO, ZN, CRIM, RM, INDUS, AGE, RAD, DIS, NOX

b. Variabile dipendente: Price

R-squared is somewhat substantial, approximately 73,5%. Thus, the model, explains almost 74% of the variability of price.

ANOVA[a]

Modello		Somma dei quadrati	gl	Media quadratica	F	Sign.
1	Regressione	31395,253	12	2616,271	113,931	<,001[b]
	Residuo	11321,042	493	22,964		
	Totale	42716,295	505			

a. Variabile dipendente: Price

b. Predittori: (costante), LSTAT, CHAS, B, PTRATIO, ZN, CRIM, RM, INDUS, AGE, RAD, DIS, NOX

The *Anova* analysis shows that F Fisher statistics have a value of 113,931, and its p-value relatively low

Therefore, chosen alpha = 0,05, the model is statistically significant.

Quindi, scelto alpha = 0,05, il modello è statisticamente significativo.

The null hypothesis according to which the regression coefficients are all zero, is refused.

Statistiche dei residui[a]

	Minimo	Massimo	Media	Deviazione std.	N
Valore previsto	-4,431	44,430	22,533	7,8847	506
Residuo	-16,1449	26,3113	,0000	4,7348	506
Valore previsto std.	-3,420	2,777	,000	1,000	506
Residuo standard	-3,369	5,491	,000	,988	506

a. Variabile dipendente: Price

Coefficienti[a]

Modello		Coefficienti non standardizzati		Coefficienti standardizzati		
		B	Errore standard	Beta	t	Sign.
1	(Costante)	34,629	5,123		6,760	<,001
	CRIM	-,107	,033	-,100	-3,216	,001
	ZN	,036	,014	,092	2,692	,007
	INDUS	-,068	,056	-,051	-1,214	,225
	CHAS	3,029	,864	,084	3,507	<,001
	NOX	-18,701	3,847	-,236	-4,862	<,001
	RM	3,912	,421	,299	9,294	<,001
	AGE	-,001	,013	-,002	-,045	,964
	DIS	-1,488	,201	-,341	-7,390	<,001
	RAD	,135	,041	,127	3,262	,001
	PTRATIO	-,985	,132	-,232	-7,478	<,001
	B	,010	,003	,095	3,521	<,001
	LSTAT	-,522	,051	-,405	-10,198	<,001

a. Variabile dipendente: Price

From the coefficients table, we can see their significance columns.

Specifically, there are some coefficients greater than 0,05.

To continue with the restricted model, variables with coefficients greater than 0,05 are eliminated, then we proceed to build the restricted model with the remaining ones.

Restricted model results

From the general model INDUS and AGE variables have been eliminated.

The remaining variables are included.

Variabili immesse/rimosse[a]

Modello	Variabili immesse	Variabili rimosse	Metodo
1	LSTAT, CHAS, B, PTRATIO, ZN, CRIM, RM, NOX, RAD, DIS[b]	.	Inserisci

a. Variabile dipendente: Price

b. Sono state immesse tutte le variabili richieste.

Riepilogo del modello[b]

Modello	R	R-quadrato	R-quadrato adattato	Errore std. della stima
1	,857[a]	,734	,729	4,7895

a. Predittori: (costante), LSTAT, CHAS, B, PTRATIO, ZN, CRIM, RM, NOX, RAD, DIS

b. Variabile dipendente: Price

As can be seen from the model summary, the R-squared is a little lower than the general model, after the elimination of some variables, whereas the adjusted R-squared has remained the same.

Thus, the variance explained from the restricted model essentially hasn't changed from the general model.

ANOVA[a]

Modello		Somma dei quadrati	gl	Media quadratica	F	Sign.
1	Regressione	31361,312	10	3136,131	136,714	<,001[b]
	Residuo	11354,983	495	22,939		
	Totale	42716,295	505			

a. Variabile dipendente: Price

b. Predittori: (costante), LSTAT, CHAS, B, PTRATIO, ZN, CRIM, RM, NOX, RAD, DIS

Coefficienti[a]

Modello		Coefficienti non standardizzati		Coefficienti standardizzati		
		B	Errore standard	Beta	t	Sign.
1	(Costante)	34,712	5,103		6,803	<,001
	CRIM	-,105	,033	-,098	-3,164	,002
	ZN	,037	,013	,093	2,731	,007
	CHAS	2,968	,861	,082	3,448	<,001
	NOX	-20,314	3,472	-,256	-5,850	<,001
	RM	3,977	,408	,304	9,754	<,001
	DIS	-1,429	,187	-,327	-7,647	<,001
	RAD	,129	,041	,122	3,157	,002
	PTRATIO	-1,015	,129	-,239	-7,867	<,001
	B	,010	,003	,096	3,591	<,001
	LSTAT	-,528	,048	-,410	-11,019	<,001

a. Variabile dipendente: Price

Statistiche dei residui[a]

	Minimo	Massimo	Media	Deviazione std.	N
Valore previsto	-4,645	44,422	22,533	7,8805	506
Residuo	-16,2609	26,2482	,0000	4,7418	506
Valore previsto std.	-3,449	2,778	,000	1,000	506
Residuo standard	-3,395	5,480	,000	,990	506

a. Variabile dipendente: Price

Model choice

We want to verify if the general model is not better than the restricted one.

We proceed to calculate the F test with the following statistical test:

$$F = \frac{\frac{[DevRes]_r - [DevRes]_i}{V_i - V_r}}{\frac{[DevRes]_i}{(n - Vi)}}$$

Here, n is the sample dimension, V_i the number of independent variables of the complete model, and V_r the number of parameters of the restricted model.
The null hypothesis is a F of Fisher with $(V_i - V_r, n - V_i)$ degrees of freedom.

The statistics has a value of:

$$F = \frac{\frac{11354{,}983 - 11321{,}042}{12 - 10}}{\frac{11321{,}042}{506 - 12}} = 0{,}740517259$$

This value is less than the critical value of 3,014 for a F of Fisher test with $\alpha = 0{,}05$ with $(2, 494)$ degrees of freedom

As a result, the null hypothesis is not refused: the restricted model is preferable

Conclusions

We now analyze the model's coefficients to say something about them.

Coefficienti[a]

Modello		Coefficienti non standardizzati B	Coefficienti non standardizzati Errore standard	Coefficienti standardizzati Beta	t	Sign.
1	(Costante)	34,712	5,103		6,803	<,001
	CRIM	-,105	,033	-,098	-3,164	,002
	ZN	,037	,013	,093	2,731	,007
	CHAS	2,968	,861	,082	3,448	<,001
	NOX	-20,314	3,472	-,256	-5,850	<,001
	RM	3,977	,408	,304	9,754	<,001
	DIS	-1,429	,187	-,327	-7,647	<,001
	RAD	,129	,041	,122	3,157	,002
	PTRATIO	-1,015	,129	-,239	-7,867	<,001
	B	,010	,003	,096	3,591	<,001
	LSTAT	-,528	,048	-,410	-11,019	<,001

a. Variabile dipendente: Price

The coefficient of the variable indicating the socio-economic status of the population (LSTAT) is negative. A sign that, all other things being equal, as the percentage of the population belonging to lower status groups in the area increases, the median home price tends to decrease. Also negative are the coefficients for crime rate (CRIM), nitrogen oxide concentration in the air (NOX), weighted distance to employment centers (DIS), and pupil-teacher ratio of the district (PTRATIO). Therefore, as the latter increase, the value of the properties would decrease (with varying degrees of intensity depending on the regressor considered).

Meanwhile, the coefficients of the remaining variables in the restricted model (such as RM, related to the average number of rooms per dwelling, and RAD, relative to accessibility to radial highways) are positive. A sign that their increase tends to raise the value of the target variable price. The value of the coefficients turns out to be moderate, with the exception of the variable RM (number of rooms), whose impact seems to be relatively high. A sign that the size and amplitude of the property's internal spaces have a significant influence on the trend of housing prices.

Residual analysis:

We examine the trend of the residuals created by the restricted model:

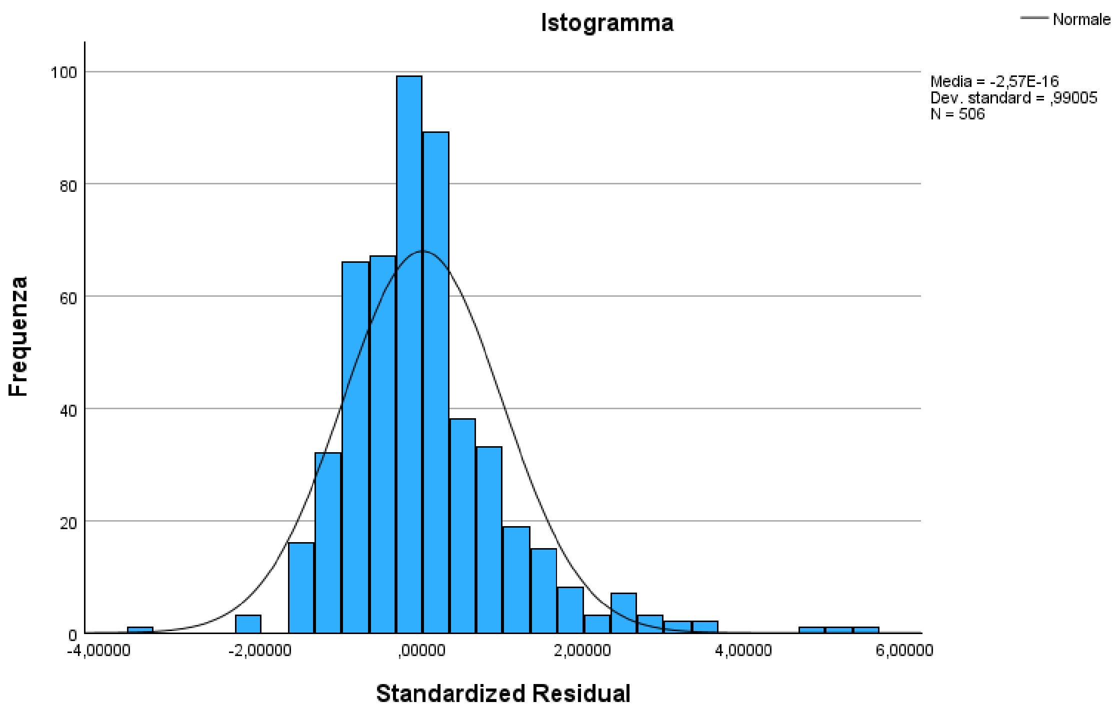
Riepilogo elaborazione casi

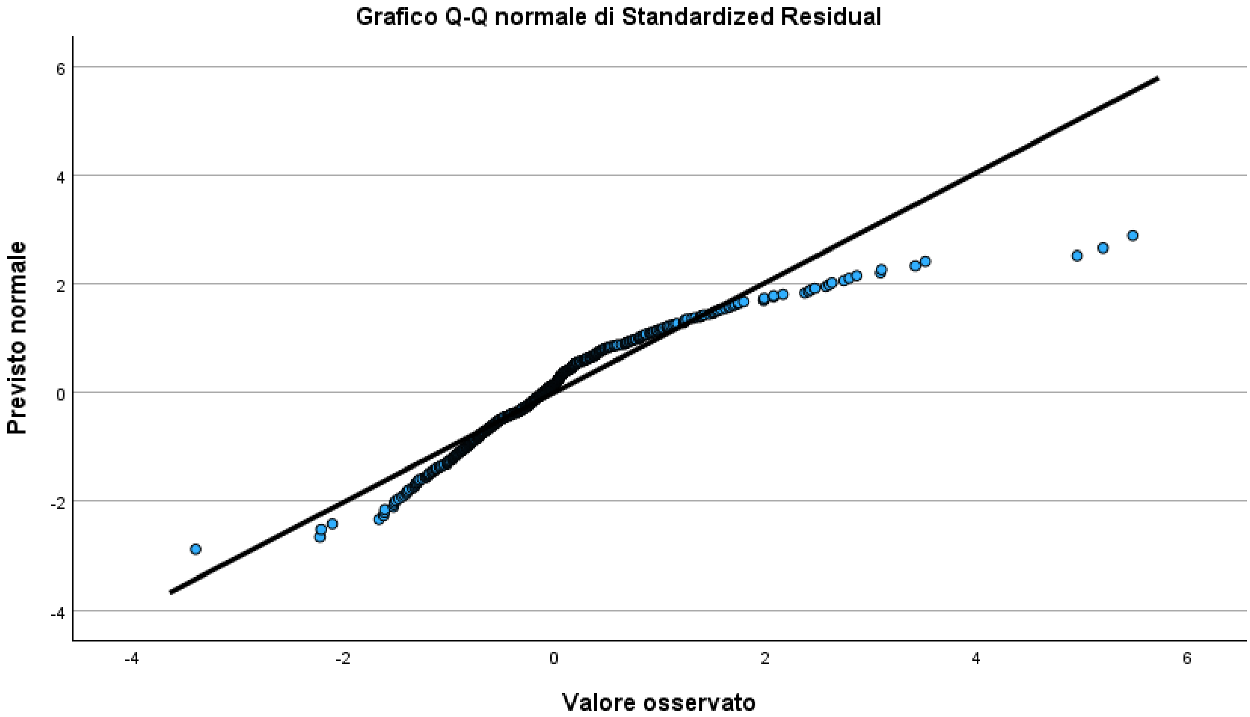
	Casi					
	Valido		Mancante		Totale	
	N	Percentuale	N	Percentuale	N	Percentuale
Standardized Residual	506	100,0%	0	0,0%	506	100,0%

Descrittive

			Statistica	Errore std.
Standardized Residual	Medio		,0000000	,04401307
	95% di intervallo di confidenza per la media	Limite inferiore	-,0864713	
		Limite superiore	,0864713	
	Media ritagliata al 5%		-,0667337	
	Mediana		-,1061274	
	Varianza		,980	
	Deviazione std.		,99004950	
	Minimo		-3,39512	
	Massimo		5,48036	
	Intervallo		8,87548	
	Intervallo interquartile		1,00401	
	Asimmetria		1,445	,109
	Curtosi		4,976	,217

The table shows that residuals has a mean of zero, they present positive asymmertry and Pronounced kurtosis, as we can see also from the graph below.





Observing the quantiles graph, we can see that the Observed residuals are placed more or less near the line.

Test di normalità

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistica	gl	Sign.	Statistica	gl	Sign.
Standardized Residual	,124	506	<,001	,908	506	<,001

a. Correzione di significatività di Lilliefors

The Shapiwo-Wilk test presents a statistical value very close to 1 of about 0,908.

Howerver, the significance is very low, this could indicate to refute the null hypothesis, excluding significantly the normality of the data distribution

Riepilogo dei casi

Riepiloghi dei casi[a]

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	Price
1	,00632	18,0	2,31	0	,5380	6,575	65,2	4,0900	1,0	296,0	15,3	396,90	4,98	24,0
2	,02731	,0	7,07	0	,4690	6,421	78,9	4,9671	2,0	242,0	17,8	396,90	9,14	21,6
3	,02729	,0	7,07	0	,4690	7,185	61,1	4,9671	2,0	242,0	17,8	392,83	4,03	34,7
4	,03237	,0	2,18	0	,4580	6,998	45,8	6,0622	3,0	222,0	18,7	394,63	2,94	33,4
5	,06905	,0	2,18	0	,4580	7,147	54,2	6,0622	3,0	222,0	18,7	396,90	5,33	36,2
6	,02985	,0	2,18	0	,4580	6,430	58,7	6,0622	3,0	222,0	18,7	394,12	5,21	28,7
7	,08829	12,5	7,87	0	,5240	6,012	66,6	5,5605	5,0	311,0	15,2	395,60	12,43	22,9
8	,14455	12,5	7,87	0	,5240	6,172	96,1	5,9505	5,0	311,0	15,2	396,90	19,15	27,1
9	,21124	12,5	7,87	0	,5240	5,631	100,0	6,0821	5,0	311,0	15,2	386,63	29,93	16,5
10	,17004	12,5	7,87	0	,5240	6,004	85,9	6,5921	5,0	311,0	15,2	386,71	17,10	18,9
11	,22489	12,5	7,87	0	,5240	6,377	94,3	6,3467	5,0	311,0	15,2	392,52	20,45	15,0
12	,11747	12,5	7,87	0	,5240	6,009	82,9	6,2267	5,0	311,0	15,2	396,90	13,27	18,9
13	,09378	12,5	7,87	0	,5240	5,889	39,0	5,4509	5,0	311,0	15,2	390,50	15,71	21,7
14	,62976	,0	8,14	0	,5380	5,949	61,8	4,7075	4,0	307,0	21,0	396,90	8,26	20,4
15	,63796	,0	8,14	0	,5380	6,096	84,5	4,4619	4,0	307,0	21,0	380,02	10,26	18,2
16	,62739	,0	8,14	0	,5380	5,834	56,5	4,4986	4,0	307,0	21,0	395,62	8,47	19,9
17	1,05393	,0	8,14	0	,5380	5,935	29,3	4,4986	4,0	307,0	21,0	386,85	6,58	23,1
18	,78420	,0	8,14	0	,5380	5,990	81,7	4,2579	4,0	307,0	21,0	386,75	14,67	17,5
19	,80271	,0	8,14	0	,5380	5,456	36,6	3,7965	4,0	307,0	21,0	288,99	11,69	20,2
20	,72580	,0	8,14	0	,5380	5,727	69,5	3,7965	4,0	307,0	21,0	390,95	11,28	18,2
21	1,25179	,0	8,14	0	,5380	5,570	98,1	3,7979	4,0	307,0	21,0	376,57	21,02	13,6
22	,85204	,0	8,14	0	,5380	5,965	89,2	4,0123	4,0	307,0	21,0	392,53	13,83	19,6
23	1,23247	,0	8,14	0	,5380	6,142	91,7	3,9769	4,0	307,0	21,0	396,90	18,72	15,2
24	,98843	,0	8,14	0	,5380	5,813	100,0	4,0952	4,0	307,0	21,0	394,54	19,88	14,5
25	,75026	,0	8,14	0	,5380	5,924	94,1	4,3996	4,0	307,0	21,0	394,33	16,30	15,6
26	,84054	,0	8,14	0	,5380	5,599	85,7	4,4546	4,0	307,0	21,0	303,42	16,51	13,9
27	,67191	,0	8,14	0	,5380	5,813	90,3	4,6820	4,0	307,0	21,0	376,88	14,81	16,6
28	,95577	,0	8,14	0	,5380	6,047	88,8	4,4534	4,0	307,0	21,0	306,38	17,28	14,8
29	,77299	,0	8,14	0	,5380	6,495	94,4	4,4547	4,0	307,0	21,0	387,94	12,80	18,4
30	1,00245	,0	8,14	0	,5380	6,674	87,3	4,2390	4,0	307,0	21,0	380,23	11,98	21,0
31	1,13081	,0	8,14	0	,5380	5,713	94,1	4,2330	4,0	307,0	21,0	360,17	22,60	12,7
32	1,35472	,0	8,14	0	,5380	6,072	100,0	4,1750	4,0	307,0	21,0	376,73	13,04	14,5
33	1,38799	,0	8,14	0	,5380	5,950	82,0	3,9900	4,0	307,0	21,0	232,60	27,71	13,2
34	1,15172	,0	8,14	0	,5380	5,701	95,0	3,7872	4,0	307,0	21,0	358,77	18,35	13,1
35	1,61282	,0	8,14	0	,5380	6,096	96,9	3,7598	4,0	307,0	21,0	248,31	20,34	13,5
36	,06417	,0	5,96	0	,4990	5,933	68,2	3,3603	5,0	279,0	19,2	396,90	9,68	18,9
37	,09744	,0	5,96	0	,4990	5,841	61,4	3,3779	5,0	279,0	19,2	377,56	11,41	20,0
38	,08014	,0	5,96	0	,4990	5,850	41,5	3,9342	5,0	279,0	19,2	396,90	8,77	21,0
39	,17505	,0	5,96	0	,4990	5,966	30,2	3,8473	5,0	279,0	19,2	393,43	10,13	24,7
40	,02763	75,0	2,95	0	,4280	6,595	21,8	5,4011	3,0	252,0	18,3	395,63	4,32	30,8
41	,03359	75,0	2,95	0	,4280	7,024	15,8	5,4011	3,0	252,0	18,3	395,62	1,98	34,9
42	,12744	,0	6,91	0	,4480	6,770	2,9	5,7209	3,0	233,0	17,9	385,41	4,84	26,6
43	,14150	,0	6,91	0	,4480	6,169	6,6	5,7209	3,0	233,0	17,9	383,37	5,81	25,3
44	,15936	,0	6,91	0	,4480	6,211	6,5	5,7209	3,0	233,0	17,9	394,46	7,44	24,7
45	,12269	,0	6,91	0	,4480	6,069	40,0	5,7209	3,0	233,0	17,9	389,39	9,55	21,2
46	,17142	,0	6,91	0	,4480	5,682	33,8	5,1004	3,0	233,0	17,9	396,90	10,21	19,3
47	,18836	,0	6,91	0	,4480	5,786	33,3	5,1004	3,0	233,0	17,9	396,90	14,15	20,0
48	,22927	,0	6,91	0	,4480	6,030	85,5	5,6894	3,0	233,0	17,9	392,74	18,80	16,6
49	,25387	,0	6,91	0	,4480	5,399	95,3	5,8700	3,0	233,0	17,9	396,90	30,81	14,4
50	,21977	,0	6,91	0	,4480	5,602	62,0	6,0877	3,0	233,0	17,9	396,90	16,20	19,4
Totale N	50	50	50	50	50	50	50	50	50	50	50	50	50	50

a. Limitato ai primi 50 casi.